Preventive Veterinary Medicine 116 (2014) 111–119

Contents lists available at ScienceDirect

Preventive Veterinary Medicine

journal homepage: www.elsevier.com/locate/prevetmed

# Comparison of time to PRRSv-stability and production losses between two exposure programs to control PRRSv in sow herds

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ARTICLE INFO

*Article history:*
Received 11 November 2013
Received in revised form 19 May 2014
Accepted 26 May 2014

*Keywords:*
PRRSv
Control
Elimination
Load-close-expose
Live virus inoculation
Modified-live virus vaccine
SPC
EWMA
Pigs

ABSTRACT

To control and eliminate porcine reproductive and respiratory syndrome virus (PRRSv) from breeding herds, some veterinarians adopt a strategy called load-close-expose which consists of interrupting replacement pig introduction for several months and exposing the pigs to a replicating PRRSv. This was a prospective quasi-experiment that followed 61 breeding herds acutely infected with PRRSv that adopted one of two exposure programs: modified-live virus (MLV) vaccine or live-resident virus inoculation (LVI). Treatment groups (load-close-expose with MLV or LVI) were compared for: (a) time-to-PRRSv stability (TTS), defined as time in weeks it took to produce PRRSv negative pigs at weaning; (b) the time-to-baseline production (TTBP), defined using statistical process control methods to represent time to recover to the number of pigs weaned per week that herds had prior to PRRSv-detection; and (c) the total production loss in terms of number of pigs weaned per week. TTS and TTBP were compared between treatments using survival analysis. Day 1 of the program was considered to be the day that treatment was administered. Sampling at herds consisted of bleeding 30 due-to-wean piglets on a monthly basis. Serum was tested for PRRSv RNA by RT-PCR. Herds in which PRRSv was not detected over a 90-day period were classified as reaching stability. Multivariate analysis using proportional hazards regression was performed adjusting the effect of treatment on TTBP and TTS to 'severity of PRRSv infection', 'number of whole-herd exposures', 'days from PRRSv-detection to intervention', 'prior PRRSv-infection status' and 'veterinary clinic associated with the herd'. Total loss was compared between groups using multivariate regression analysis adjusted by selected covariates. The median TTS among participating herds was 26.6 weeks (25th to 75th percentile, 21.6–33.0 weeks). The overall TTBP was 16.5 weeks (range 0–29 weeks). The magnitude of production losses following whole-herd exposure averaged 2217 pigs not weaned/1000 sows and was correlated with TTBP. Herds in the MLV group recovered production sooner and had less total loss than herds in the LVI group. TTBP and TTS were significantly shorter and the total loss was significantly less in herds assisted by a specific veterinary clinic and herds that were infected with PRRSv in the 3 years prior to the study. This study provided new metrics to assist veterinarians to decide between methods of exposure to control and eliminate PRRSv from breeding herds.

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http://dx.doi.org/10.1016/j.prevetmed.2014.05.010
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## 1. Introduction

Since its first report in the US swine industry 23 years ago (Keffaber, 1989), porcine reproductive and respiratory syndrome virus (PRRSv) continues to cause significant pig production losses in North America and around the world (Lunney et al., 2010; Zimmerman et al., 2012). The annual cost of porcine reproductive and respiratory syndrome virus (PRRSv) to the US swine industry was estimated at 664 million dollars (NPB, 2011). To reduce production impact, different strategies to control and eliminate PRRSv at herd and regional levels have been employed (Corzo et al., 2010; Rowland and Morrison, 2012).

A standardized terminology to communicate PRRSv-status for breeding herds has been proposed (Holtkamp et al., 2011). Briefly, PRRSv-infected herds can be classified as "positive unstable". As PRRSv is controlled and ultimately eliminated, herds move to "positive stable" status, then to "provisional negative" and ultimately to "negative" status. Because the stable herd status is based on statistical sampling, PRRSv can be detected in some stable herds, but frequency will be low. Provisional negative herds must have sentinel gilts remain PRRSv negative for at least 60 days thereby building confidence that virus is not present. All pigs from "negative" herds are negative for anti-PRRSv antibodies (Holtkamp et al., 2011).

In North America, herd closure combined with whole-herd exposure to a live PRRSv is a commonly attempted strategy to control and eliminate the virus from infected breeding herds (Yeske, 2009). The concept of herd closure is that pig introduction into breeding herds is interrupted for a period of time to allow the pathogen to die out. The underlying principle is to prevent the introduction of susceptible pigs that, in contact with PRRSv, could get infected and disseminate virus within the herd perpetuating the infection (Torremorell et al., 2003). The strategy of whole-herd exposure was developed as an addition to the traditional herd closure approach and has the principle of exposing all pigs in the herd with a live replicating PRRSv (Desrosiers and Boutin, 2002). As a consequence, it is expected that most pigs will get infected, become infectious in the short term and immune in the long term thereby homogenizing the immune status of animals in the population. When herd closure is combined with whole-herd exposure (strategy called load-close-expose), the expectation is that protective immunity is built at the population level resulting in the decrease and eventual elimination of PRRSv from the breeding herd. Consequently, newborn piglets do not become infected and thus are weaned "PRRSv negative" (Holtkamp et al., 2011).

Two immunogens commonly used for PRRSv whole-herd exposure are PRRSv modified-live virus (MLV) vaccine and the live-resident herd virus. However, it is unknown which protocol leads to a lesser production impact in the breeding herd. The objectives of this study were to compare: (a) the time required to PRRSv-stability (TTS) between breeding herds that used MLV or live-resident virus inoculation (LVI) as immunogens as part of the load-close-expose programs, (b) the time it took for farms to recover the production levels they had prior to PRRSv infection (time to baseline production, TTBP) and (c) the magnitude of the production losses following PRRSv infection and adoption of load-close-expose strategy using LVI or MLV. Productivity was measured as "total number of pigs weaned per 1000 sows" per week (Schaefer and Morrison, 2007).

To the best of our knowledge, this is the first study to use statistical process control methods to define return to "in control" levels of livestock production parameters. The methods described in this manuscript can be used in future studies to evaluate effectiveness of different preventive or therapeutic strategies on production impact of diseases.

## 2. Design and methods

### 2.1. Study design

We conducted a prospective quasi-experimental study enrolling breeding pig herds that were infected with PRRSv and adopted load-close-expose program with either MLV vaccine (treatment A) or LVI inoculation (treatment B) as part of PRRSv elimination program.

Herds were assumed to achieve "TTS status" when there was a failure to detect PRRSv RNA in serum of pre-weaning pigs by RT-PCR tested monthly over a 90-day period. Day 1 of the program was the day that the herd was exposed to either MLV or LVI.

Change in 'number of pigs weaned per 1000 sows per week' was compared between herds that used MLV vaccine to that of herds that used LVI. To describe the changes in number of pigs weaned, we developed 3 metrics, which are described in Table 1.

### 2.2. Source population

Veterinarians listed in the American Association of Swine Veterinarians (AASV) directory were invited via email to enroll herds to which they provided technical services. Eligible herds were those acutely infected with PRRSv from November 2009 to March 2012 and that adopted load-close-expose program with intent to produce PRRSv-negative pigs at weaning. Enrolled herds fulfilled the following inclusion criteria: (a) stable sow inventory (herd not expanding or shrinking by more than 10%); (b) diagnostic evidence that PRRSv was present; (c) intent of herd owner to eliminate PRRSv from the herd; (d) no replacement pigs introduced during the closure period; (e) owners willing to share production performance data, (f) enrollment in the study within 2 months of starting the load-close-expose program and (g) enrollment in the study before the PRRSv monitoring phase started.

Herds were dropped from the study if (a) the herd veterinarian indicated that there was a new unrelated PRRSv introduction, (b) the participating herd had introduced replacement animals before TTS was reached or (c) the owner opted to withdraw.

A total of 61 herds were enrolled and information on herd demographics, type of PRRSv strain and PRRSv infection history of the herd were recorded (Table 2). From the 61 herds, 3 were kept out of the production analyses (TTBP and loss calculation) because data was not available for those herds (Table 3).

**Table 1**
Parameters used to characterize change in total pigs weaned per week.

| Parameter | Meaning | Method of calculation |
|---|---|---|
| Time to baseline production (TTBP) | Time, in weeks, it took to recover the levels of 'weaned pigs per week' that the herd had prior to PRRSv-detection (i.e. time to "in control" levels of productivity[a]). | Defined by statistical process control using the exponentially weighted moving average (EWMA) control chart method (Montgomery, 2012).[b] |
| "Δ pigs A" | Total number of pigs not weaned/1000 sows from the time of PRRSv-detection to time of herd exposure to LVI or MLV. It indicated the severity of PRRSv break and was used to adjust comparisons of the "Δ pigs B" parameter. | Calculated by adding the sums of total number of pigs weaned below the expected value (21 week-average before PRRSv-infection) from time of PRRSv-detection until whole-herd exposure to either LVI or MLV. |
| "Δ pigs B" | Total number of pigs not weaned/1000 sows from the time of whole-herd exposure to TTBP status achievement. It represented the 'total production loss' following PRRSv infection and load-close-expose using LVI or MLV. | Cumulative sum of number of pigs weaned below the expected from the time of whole-herd exposure to TTBP status. |

[a] For the purpose of this study, productivity was measured as quantity of pigs weaned per week.
[b] Details on the parameters used to construct EWMA are provided in Section 2.6.

**Table 2**
Baseline demographic characteristics of enrolled herds in the study for the TTBP and total loss outcomes, before establishment of load-close-expose as part of PRRSv elimination programs.

| Characteristic | LVI | MLV |
|---|---|---|
| Number enrolled | 40 | 18 |
| Prior infection | 20 (50%) | 12 (65%) |
| RFLP strain 1-4-4 | 15 (38%) | 10 (55%) |
| Herd size (mean ± SE) | 3400 ± 303 | 2471 ± 250* |
| Days from PRRSv detection to intervention (mean ± SE) | 28 ± 3 | 39 ± 9 |

* Difference between groups was statistically different at alpha level of 0.05.

The study was approved by the University of Minnesota's IACUC (protocol number 1004A80315) and all the farm owners or their representatives provided written, informed consent agreeing to participate.

### 2.3. PRRSv used for breeding herd exposure

Herds in the MLV group were vaccinated with either Ingelvac® PRRS ATP (Boehringer Ingelheim Vetmedica Inc., St. Joseph, MO) vaccine or Ingelvac® PRRS MLV (Boehringer Ingelheim Vetmedica Inc., St. Joseph, MO) vaccine. LVI protocols were based on application of diluted PRRSv collected from serum from infected pigs from the farm (Batista et al., 2002).

**Table 3**
Baseline demographic characteristics of the herds enrolled in the study for the TTS outcome, before establishment of load-close-expose as part of PRRSv elimination programs.

| Characteristic | LVI | MLV |
|---|---|---|
| Number enrolled | 41 | 20 |
| Prior infection | 20 (49%)[a] | 13 (65%)[a] |
| RFLP strain 1-4-4 | 15 (38%)[a] | 12 (60%)[a] |
| Herd size (mean ± SE) | 3400 ± 303[a] | 2471 ± 250 |
| Days from PRRSv detection to intervention (mean ± SE) | 28 ± 3[a] | 39 ± 9[a] |

[a] There were no significant differences between groups at alpha level of 0.05 for prior infection and RFLP 1-4-4 (Fisher's exact, *p*-values 0.2813 and 0.1102, respectively), and for days from PRRSv-detection to intervention (*t*-test, *p*-value 0.2526). For herd size, LVI herds were significantly larger than MLVs (*t*-test *p*-value 0.0324). This sample size allowed a detection of a hazard ratio (to achieve TTS) of at least 2.3 between treatment groups with 80% statistical power.

### 2.4. PRRSv monitoring

PRRSv-monitoring started 12 weeks after day 1 and consisted of bleeding 30 pre-weaned piglets (determined by target prevalence to detect at least one positive assuming at least 10% prevalence at a 95% confidence level for any population size) and testing serum samples for PRRSv-RNA by RT-PCR. This was the scheme suggested by the American Association of Swine Veterinarians to monitor a breeding herd to PRRSv and to define herd as "stable" (Holtkamp et al., 2011). Serum samples were submitted to a reference veterinary diagnostic laboratory for testing in pools of 5 (Rovira et al., 2007) following the standardized procedures for each laboratory.

Farms that obtained intermittent PCR-positive results (e.g. two months of negative results followed by a positive result) submitted the serum samples for PRRSv RNA open reading frame 5 (ORF 5) molecular sequencing. The sequences were compared to the resident virus (phylogeny comparison of the PRRSv-ORF5 previously detected in the herd). For the purpose of this study, unexpected PCR-results were considered to be the same resident virus (no new PRRSv introduction) when there was (a) no apparent increase in clinical signs, and (b) the sequenced ORF 5 was at least 97% similar to the resident virus present at the beginning of the load-close-expose program (Alonso, 2012; Murtaugh, 2012).

### 2.5. Statistical analysis for the TTS outcome

The effect of multiple covariates on TTS and their association with the main effect (MLV or LVI treatment) was assessed. The following covariates were considered in the analysis: herd size, previous PRRSv infection, PRRSv restriction fragment length polymorphism (RFLP) pattern 1-4-4, time between PRRSv-detection and whole-herd exposure and veterinary clinic that provided technical services.

Herds that were PRRSv-free (AASV category IV) (Holtkamp et al., 2011) for the previous 3 years before PRRSv infection were considered to have "no prior PRRSv infection". Herds with at least one PRRSv introduction in the past 3 years before the current infection were

considered to have "PRRSv prior infection". A dichotomous variable "RFLP pattern 1-4-4" was created to adjust analysis for classifying PRRSv belonging to a similar virus genotyping group (Shi et al., 2010).

Power calculation indicated that a minimum of 31 and 15 herds for the LVI and MLV groups respectively were needed to have at least 80% power to detect differences between treatments of a hazard ratio of at least 2.50, with the use of a log-rank test at a two-sided significance level of 0.05. Overall, a higher number of herds in the LVI group were enrolled because the number of eligible herds using MLV vaccine in load-close-expose programs was proportionally fewer than that of LVI-herds.

Treatment groups (MLV or LVI) were compared with the use of Kaplan–Meier survival estimates for TTS, with censoring data for herds that dropped from the study after PRRSv monitoring started (12 weeks) and before reaching TTS. Herds that dropped from the study before starting PRRSv monitoring were excluded from the analysis. Between group differences in survival curves were assessed with the use of log-rank test.

Cox proportional-hazards regression was used to calculate the magnitude of the treatment effect on TTS and to adjust the analysis for the aforementioned covariates. The proportional hazards assumption for the proportional hazards models was checked with the Kolmogorov test and by examining the plot of the natural logarithm of cumulative hazard versus time, with the expectation that the plots would be parallel. Factors associated with TTS with a *p*-value less than 0.20 in the univariate analysis were entered in the multivariate model and non-significant factors were removed by means of a backward-selection procedure. All non-significant variables that were eliminated were re-entered into the model one by one to ensure no exclusion of significant or confounding variables. It was checked for interaction effects between the main effect and other covariates. Two-sided hypotheses and tests were used for all statistical inferences. A *p*-value of less than 0.05 was considered to indicate statistical significance and the goodness-of-fit test was tested using the Hosmer and Lemeshow test.

Statistical analyses were performed with the use of SAS 9.2 software (SAS Institute Inc., Cary, North Carolina).

### 2.6. Statistical analysis for the TTBP and total loss outcomes

Exponentially weighted moving average (EWMA) was the statistical process control tool constructed to define TTBP. Chart parameters were calibrated based on the intention of operating with a low false-alarm rate (i.e. detect 'back to *in control*' with high specificity) and thus it was used weight ($\lambda$) of 0.40 and a multiple of sigma ($\sigma$) of 3 for control limits (Krieter et al., 2009). Such EWMA parameters resulted in an average time to signal of 1.97 weeks, meaning that in the long run, the control chart required no more than approximately 2 weeks to detect TTBP.

TTBP was compared between groups with the use of Kaplan–Meier survival estimates. Differences in survival curves between groups were assessed with the use of log-rank test. Moreover, Cox proportional-hazards regression was used to calculate the magnitude of the treatment effect on TTBP and to adjust analysis for covariates. The following covariates were considered in the analysis: '$\Delta$ pigs A' (delta pigs A), 'number of whole-herd exposures', 'days from PRRSv-detection to intervention', 'prior infection status' and 'veterinary clinic responsible for technical assistance for the herd'. Factors associated with TTBP with a *p*-value less than 0.20 in the univariate analysis were entered in the multivariate model and non-significant factors were removed by means of a backward-selection procedure. All non-significant variables that were eliminated were re-entered into the model one by one to ensure no exclusion of significant or confounding variables. It was checked for interaction effects between the main effect and other covariates

The outcome '$\Delta$ pigs B' (delta pigs B) was compared between treatment groups using *t*-test. General linear models were constructed to adjust the effect of treatment on '$\Delta$ pigs B' by the same covariates considered in the TTBP analysis.

Herds that were PRRSv-free (AASV category IV) (Holtkamp et al., 2011) during the previous 3 years prior to the present study were considered to have "no prior PRRSv infection". Herds with at least one PRRSv introduction in the past 3 years before the study were considered to have "PRRSv prior infection".

Spearman correlations were calculated between TTS, TTBP, "$\Delta$ pigs A", "$\Delta$ pigs B" and "days from PRRSv-detection to load-close-expose".

Two-sided hypotheses and tests were used for all statistical inferences. A *p*-value of less than 0.05 was considered to indicate statistical significance. Statistical analyses were performed using the SAS 9.2 package (SAS Institute, Inc., Cary, NC).

## 3. Results

### 3.1. Descriptive analysis, TTS outcome

Eighty-five percent of herds in the LVI program and 60% of herds in the MLV program reached stability by the end of the observational period. One herd in the LVI and 3 herds in the MLV group did not reach TTS by the end of the observational study period, 5 herds (3 LVI and 2 MLV) dropped out from the study because of detection of a PRRSv with <97% homology compared to the virus identified at the beginning of the load-close-expose program, 4 herds (2 LVI and 2 MLV) dropped out because the producer decided to re-start the load-close-expose program and 1 MLV herd discontinued because it caught on fire and needed to be depopulated.

TTS ranged from 12 to 42 weeks among 61 participating herds with a median of 26.6 weeks (25th to 75th percentile, 21.6–33.0 weeks). From 61 farms, 24 had at least 1 month of PRRSv PCR-negative results followed by positive PCR results (Table 4). Overall, frequency of having at least one PCR negative result followed by a positive one tended to be higher in the MLV group (Chi-square *p*-value 0.0804).

In contrast to MLV group, the LVI group had no events of 2 or 3 negative tests followed by a positive result (Table 6). Farms that had at least two consecutive PCR-negative followed by PCR-positive results obtained sequences at least

**Table 4**
Number of herds (and percentage of total herds) by treatment group with intermittent pattern of PRRSv RNA detection by RT-PCR according to detection pattern.

| Pattern of PRRSv RNA by RT-PCR detection | LVI | MLV |
|---|---|---|
| Straight 4 consecutive negative monthly tests (TTS) (pattern:+ − − − −)[a] | 28 (68%) | 9 (45%) |
| One negative monthly test followed by a positive test (pattern: + − +) | 13 (32%) | 7 (35%) |
| Two negative monthly tests followed by a positive test (pattern + − − +) | 0 (0%) | 2 (10%) |
| Three negative monthly tests followed by a positive test (pattern + − − − +) | 0 (0%) | 2 (10%) |

[a] The pattern in parenthesis represents monthly PRRSv-RNA PCR results: the plus sign (+) represents a month with at least 1 PCR-positive result. Negative sign (−) represents a month with all PCR results being negative for PRRSv-RNA.

97% similar to the original resident virus, suggesting that those unexpected PCR-positive were not explained by lateral PRRSv infection (Christopher-Hennings et al., 2002; Murtaugh, 2012).

### 3.2. Descriptive analysis, TTBP and total loss outcomes

The overall median TTBP was 16.5 weeks and ranged from 0 to 29 weeks. The magnitude of production losses following whole-herd exposure (Δ pigs B) ranged from 0.0 to 9561.0 pigs/1000 inventoried sows and averaged 2217.2 pigs not weaned/1000 sows.

The average number of pigs not weaned from PRRSv detection to exposure (Δ pigs A) was 571.9 pigs/1000 sows. The Δ pigs A parameter was slightly higher in the LVI group ($p = 0.0681$) (Table 5).

TTBP had a Spearman correlation coefficient of 0.406 ($p$ 0.0016) and 0.885 ($p < 0.0001$) with Δ pigs A and Δ pigs B, respectively. The correlation between TTBP and TTS was 0.594 ($p = 0.0002$) for herds in the LVI group and 0.0850 ($p = 0.7557$) in the MLV group. The correlation of TTBP and time from PRRSv-detection to treatment was −0.201 ($p = 0.1305$).

### 3.3. Univariate analysis, TTS outcome

Herds that were exposed with LVI had significantly shorter TTS than herds treated with MLV (Log rank $p$-value 0.0137) (Fig. 1A). Similarly, herds with prior PRRSv infection reached TTS significantly sooner than herds without PRRSv prior-infection (Log rank $p$-value 0.0275) (Fig. 1B).

**Table 6**
Adjusted effects of treatment and other covariates on TTBP (proportional hazards regression analysis).

| Parameter | Hazard ratio | $p$-Value* |
|---|---|---|
| Treatment (LVI vs. MLV) | 3.59 | 0.0005 |
| Δ pigs A | 0.99 | 0.1000 |
| Prior infection status (yes vs. no) | 2.50 | 0.0031 |
| Veterinary clinic | 3.39 | 0.0007 |

* Chi-square $p$-values from Type III tests of Cox proportional hazards regression.

RFLP pattern 1-4-4, herd size and time from PRRSv-detection to whole-herd exposure were not statistically associated with TTS (Log-rank $p$-values 0.5306, 0.4455 and 0.5671, respectively).

### 3.4. Univariate analysis, TTBP and total loss outcomes

Herds in the MLV group recovered production sooner ($p < 0.0001$) and had less total loss ($p = 0.0095$) than herds in the LVI group (Table 6). Similarly, herds assisted by veterinarians from a specific veterinary clinic reached TTBP sooner and had less production losses due to PRRSv than herds assisted by other veterinarians ($p = 0.0031$ and $p = 0.0109$, respectively). Herds 'with' and 'without' prior PRRSv infection had a median TTBP of 11 weeks (25th to 75th percentile: 5–20) and 21 weeks (25th to 75th percentile: 16–24) ($p = 0.0014$), respectively. Herds with prior PRRSv infection had 1907 pigs/1000 sows less total loss than herds without prior PRRSv infection ($p = 0.0001$).

### 3.5. Multivariate analysis, TTS outcome

The variables that remained in the final multivariate model were treatment (LVI or MLV) and prior PRRSv-infection (yes or no). The following variables were not associated with TTS: herd size, RFLP pattern and days from PRRSv-detection to whole-herd exposure.

After adjusting for treatment and prior PRRSv-infection status, herds assisted by a specific veterinary clinic achieved TTS significantly sooner than all other herds ($p$-value 0.0429).

The final Cox regression model hazard ratio for TTS was 2.9 for LVI herds compared to MLV herds ($p$-value 0.0034) and 2.5 for herds with prior PRRSv infection compared to herds without prior-PRRSv infection ($p$-value 0.0036).

### 3.6. Multivariate analysis, TTBP and total loss outcomes

The proportional hazards regression model indicated that all variables significant in the univariate analysis maintained significance after being adjusted by each other (Table 6).

**Table 5**
Comparison of PRRSv severity (Δ pigs A), total loss (Δ pigs B) and time to baseline production (TTBP) between groups (univariate analysis).

| | LVI | MLV | $p$-Value* |
|---|---|---|---|
| Δ pigs A (mean ± std. error) | 678.4 ± 106.0 | 335.3 ± 141.4 | 0.0681 |
| Δ pigs B (mean ± std. error) | 2665.0 ± 313.0 | 1222.2 ± 395.3 | 0.0095 |
| TTBP (median and 25th to 75th percentile) | 21(13, 24) | 10(0, 15) | <0.0001 |

* Between group comparisons for Δ pigs A and Δ pigs B were done using $t$-test and for TTBP using Log-rank test.

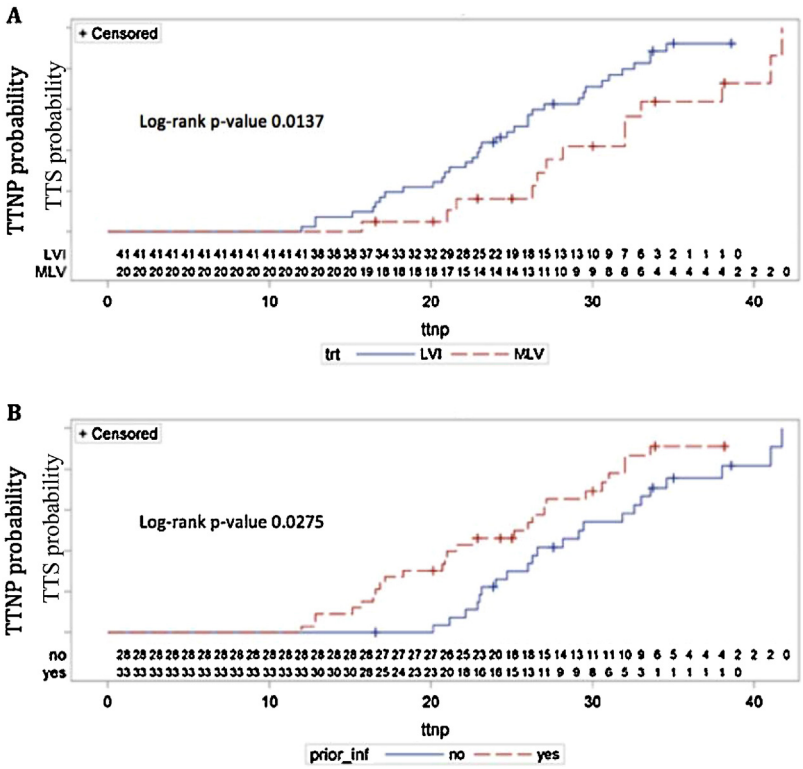


**Fig. 1.** Kaplan–Meier estimates for time-to-PRRSv-stability (TTS) according to whole-herd exposure treatment (A) and herd status prior to PRRSv-infection (B). The log-rank *p*-values included in each plot tested the whole survival curves and mean that TTS was significantly different according to (a) treatment type or (b) prior PRRSv infection status. *Panel A*: Median TTS and interquartile range (25th to 75th percentiles) for herds treated with LVI and MLV were 25.1 weeks (20.7, 31.0) and 32.0 weeks (26.6, 41.0), respectively. *Panel B*: Median TTS and interquartile range for herds with and without prior PRRSv-infection history were 26.0 weeks (17.1–31.0 weeks) and 29.3 weeks (24.0–34.6 weeks), respectively.

Similarly, a multivariate general linear model that adjusted the effect of Δ pigs A, treatment, prior infection status and veterinary clinic resulted in all the parameters remaining significant predictors of Δ pigs B after adjustment by each other (*p*-values = 0.0114, 0.0285, 0.0132 and 0.0040, respectively).

## 4. Discussion

This study was a prospective quasi-experiment study where the effectiveness of "PRRSv herd closure" programs was evaluated using 3 different outcomes. Breeding herds acutely infected with PRRSv that adopted load-close-expose programs with either LVI or MLV with intent to produce PRRSv negative pigs at weaning were followed. First, we reported TTS between groups. TTS can be defined as a proxy of time to produce PRRSv-negative pigs at weaning. Furthermore, we compared the impact of the exposure method on the time it took to achieve baseline production (TTBP) and on the magnitude of the production losses associated with PRRSv infection. Outcomes of this project will help veterinarians make informed decisions on which load-close-expose strategy to use to control and eliminate PRRSv from breeding herds.

Overall our results showed a wide variation of the investigated outcomes (TTS, TTBP, total loss). The median TTS of 26.6 weeks was longer than we anticipatedgiven previous reports in the literature (Yeske, 2009; Torremorell, 2011; Rowland and Morrison, 2012; Zimmerman et al., 2012). Perhaps the previous reports were based on a few "success" cases and did not fully represent the range of herds represented in this study. To our knowledge, this is the first study to prospectively follow a relatively high number of PRRSv infected herds attempting to eliminate the virus and thus we speculate that the median TTS reported in this study could be extrapolated to other herds that have characteristics similar to the herds enrolled in this study.

The adjusted comparisons showed that after adjusting for prior PRRSv infection status, herds of the LVI group had a higher TTS hazard ratio than those of MLV group. Given that a particular herd did not reach TTS yet, it had a 2.9 relative risk (hazard ratio) of reaching TTS in the following week.

There are two fundamental differences between exposure programs that could explain why LVI herds achieved TTS faster than MLV herds. First, immune response to virulent PRRSv is faster and stronger than that to attenuated virus, as measured by duration and magnitude of viremia and antibody titers (Klinge et al., 2009). Secondly, even though the degree of RNA nucleotide similarities between viruses does not predict degree of immunological cross-protection (Prieto et al., 2008; Murtaugh and Genzow, 2011), exposing the breeding herd to resident virus may induce quicker immune response than that induced by a non-related virus such as the one found in the MLV vaccine (Cano et al., 2007a,b).

Whole-herd exposure with attenuated virus vaccine has the advantage of potentially causing fewer clinical signs after exposure and potentially, lower the risk of regional field virus dissemination (Linhares et al., 2012). Further studies to compare effectiveness of LVI vs. MLV in regards to quantifying the production impact are needed.

During the period of data collection, veterinarians from participating herds reported their perception that "PRRS viruses of RFLP pattern 1-4-4 tended to persist longer in infected populations compared to those of other RFLP patterns". About half of the enrolled herds were infected with RFLP pattern 1-4-4. However, both univariate and multivariate analyses did not reveal an association between infection with PRRSv RLFP pattern 1-4-4 and TTS. Further studies using more discriminatory molecular techniques to differentiate PRRSv isolates could reveal RNA segments associated with TTS.

Based on an annual herd replacement rate of 40–50%, we assumed that herds that had not detected PRRSv infection for the last 3 years had little to no remaining herd immunity to PRRSv. Thus, we created a dichotomous variable to differentiate herds in regards to "prior PRRSv infection in the past 3 years". The results showed that herds with prior infection achieved TTS before than herds that had not reported prior infection (low immunity herds). Prior infected pigs develop immunity to PRRSv faster than pigs infected for the first time (Murtaugh et al., 2002). Therefore, we speculate that a similar phenomenon happens at the herd level. Prior immunity present in herds with prior infection may explain the fact that those herds achieved TTS and TTBP first due to the development of adaptive immune response present in previously exposed pigs.

Assuming that "prior PRRSv herd immunity" was the reason why herds with "prior PRRSv infection" achieved TTS and TTBP sooner than herds without prior PRRSv infection (or herd immunity), we highlight the need of further studies to better describe how long PRRSv immunity lasts at a population level and, similarly, what is the minimum period of time necessary for the protective effect of "prior infection status" on TTS and TTBP. We speculate that factors that affect anti-PRRSv herd immunity level such as PRRSv immune status of replacement animals and breeding herd replacement rate play an important role on the effect of herd immunity on TTS and TTBP.

However, it has been reported that prior exposure to pathogens can also result in an epidemic that is longer in duration than if the pathogen was introduced into a naïve population (Pulliam et al., 2007). We speculate that this phenomenon called "epidemic enhancement" did not happen in the studied populations because in this study the whole herd was exposed to the virus at once and herds were closed to the introduction of PRRSv-susceptible replacement pigs for a period of time sufficient to significantly decrease the risk of PRRSv-shedding and transmission events within the herd.

"Time to PRRSv-stability" in the pre-weaned pig population is an important parameter to assist producers to make informed decision in regards to which protocol to use to control and eliminate PRRSv from breeding herds. However, other important aspects must be considered when deciding between protocols, including (a) if the protocol has an effect on the clinical impact, and (b) the risk of exposing neighbors to new virus isolates.

The correlation between TTBP and TTS depended on the treatment and was weak for MLV and moderate for LVI (Taylor, 1990). This suggests that productivity of weaned pigs was a poor indicator of the PRRSv status of the herd.

The correlation coefficients between TTBP and Δ pigs A and Δ pigs B were weak and strong, respectively (Taylor, 1990) meaning that the severity of the initial production loss could not predict TTBP. However, the higher the total loss following load-close-expose, the longer it took for herds to recover productivity. Some herds had no statistically significant change in productivity following PRRSv infection and adoption of load-close-expose programs, while other herds required a relatively long time to recover production levels (reach TTBP) and had severe production losses (as high as 9561 pigs/1000 sows). Our study differed from a previous report where PRRSv elimination from a breeding herd using load-close-expose with MLV vaccine had no impact on number of pigs weaned (Voglmayr et al., 2006).

A study done by Schaefer and Morrison comparing total number of pigs weaned 52 weeks before and 52 weeks after herd closure implementation to eliminate PRRSv from infected breeding herds found that 13 out of 15 herds weaned similar or more pigs after the herd closure program (Schaefer and Morrison, 2007). The main differences between that study and the present one are that breeding herds in this study were acutely infected with PRRSv and adopted load-close-expose whereas in Schaefer and Morrison's study herds were endemically infected and used herd closure only (no deliberated exposure to live PRRSv).

Herds in the MLV group had numerically smaller Δ pigs A (delta pigs A) than LVI herds indicating that severity of PRRSv infection at the moment of load-close-expose was less for MLV herds. Severity of PRRSv break may have been one of the factors that influenced veterinarian's decision to choose between MLV or LVI as immunogen for load-close-expose programs. Therefore, Δ pigs A was included as a covariate in the multivariate regression models to compare TTBP and Δ pigs B between treatment groups, adjusting effects to that variable.

Data from this study proves that during PRRSv elimination programs PRRSv monitoring must be done repeatedly over time (Table 5), as previously suggested (Holtkamp et al., 2011). We showed that some herds in the MLV program had at least one PRRSv RNA RT-PCR negative test followed by positive result. This might be explained by MLV herds recovering productivity sooner than LVI herds, suggesting that PRRSv infection prevalence was lower in the MLV group compared to LVI group.

We speculate that because production levels recovered sooner than TTS in MLV herds, farm personnel might have assumed that PRRSv was already eliminated from the herd. Thus, extra management practices adopted at the herd to limit within farm transmission of PRRSv (such as cross fostering of piglets, washing and down time of farrowing crates) were relaxed sooner than achievement of PRRSv negative status, favoring re-circulation of PRRSv and contributing to a longer TTS for MLV herds.

The fact that herds assisted by a specific veterinary clinic reached TTS and TTBP sooner and had a less severe total loss might be explained by specific standard operating procedures followed by such herds during herd closure. Therefore there is a need to study the relationship between management practices and herd closure success.

Statistical process control (SPC) charts have been used in veterinary medicine to monitor changes in key performance indicators related to livestock production (De Vries and Reneau, 2010). There are different types of SPC charts and the most commonly described are Shewhart and EWMA (De Vries and Reneau, 2010). The advantages of EWMA over Shewhart include that EWMA has fewer false alarms, is more flexible allowing adjustments for better sensitivity or better specificity, and there is no need for extra sensitizing rules (Montgomery, 2012). Also, when EWMA is used for individual measurements, which was the case in this study (number of pigs weaned per week), it is considered a distribution free (nonparametric) procedure (Montgomery, 2012). In the swine industry, EWMA has been used to identify small deviations in repeat estrus rate and total born per litter (Krieter et al., 2009). However, this is the first report of a SPC tool to detect "back to control" state of a production parameter.

Strengths of this study include the nature of the investigation which was an applied quasi-experimental study performed with a relatively high number of herds under field conditions, with well-defined inclusion criteria.

Some of the limitations of this study are that (a) monitoring duration and frequency may have missed some infected herds, (b) the existence of unknown confounder variables might have biased the findings and (c) the allocation of treatment was not random (quasi-experiment). Moreover, participating herds did not submit samples to the same diagnostic laboratory. Therefore, differences in PCR sensitivity might have influenced results (Gerber et al., 2012). Moreover, to calculate TTBP the baseline levels were arbitrarily set to the average of 21 weeks prior to PRRSv-detection. Analysis with herds from this study that were PRRSv-negative (AASV category IV) (Holtkamp et al., 2011) showed a high correlation between 52 and 21 week-average in the total number of weaned pigs (data not shown). We chose 21 weeks because it represents a full breeding cycle and because not all herds had stable sow inventory during 52 weeks prior to the study.

This study provided important metrics on load-close-expose projects using LVI or MLV as methods of whole-herd exposure. Time it takes to reach PRRSv stable herd, time it takes to recover production levels that the herds had before PRRSv infection and number of pigs "not" weaned were described. Further studies are needed to elucidate the effect of management practices on herd closure success as well as to develop economic models using information here described to assist in the decision between using LVI or MLV as part of load-close-expose programs to control and eliminate PRRSv from breeding herds.

## 5. Conclusions

Farms that fulfilled the eligibility criteria of this study achieved TTS sooner if breeding animals had prior contact with PRRSv and when LVI was the method of whole-herd exposure. Herds that used MLV vaccine as the method for whole-herd exposure had significantly shorter TTBP and less severe production losses compared to farms that used LVI. Similarly, TTS and TTBP were significantly shorter and productivity was less affected in farms with prior PRRSv infection and herds assisted by a specific veterinary clinic. Moreover, the correlation between TTBP and TTS was significant for herds using LVI but not significant for herds in the MLV program, as herds in the MLV had a variable and often longer period from TTBP to TTS status compared to LVI group. Further studies are needed to better understand the association of management practices with herd closure success.

## Acknowledgements

This study received funds from University of Minnesota Swine Disease Eradication Center (SDEC) and from Boehringer Ingelheim Vetmedica Inc.

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